Flat plate drive

A flat plate drive is a method of turning a bowl to a finish between centres and involves the bowl face being pressed against a plywood or timber disc by the tailstock. To soften the contact area we are going to glue on a layer of router matting which will create a good contact pad that will protect the surface of your bowl and provide a good non-slip surface for the drive plate at the same time. Again, I prefer to use a good-quality birch plywood and 25mm is a must; much thinner and the ply will have too much bend in it.

As well as a flat plate there is nothing stopping you turning a slightly radius plate drive from a bowl blank and placing some router matting between this and the bowl before turning. It's important that you add some type of cushioning between the bowl and drive to avoid damage to the bowl. It's also worth spending a bit of time making these plates as they will give you many years of reliable service. Here's how to do it...

1 To start making a flat plate drive you will need to decide on the size of plate you want. This will depend on the size of bowls that you most commonly make. Here, I'm making a 360mm plate which will do a wide variety of bowls and platters. I'm mounting a face plate ring onto the disc which will mean I can easily and quickly mount the plate onto my chuck after taking the bowl off

2 On the other side of the plate I'm using a contact adhesive to stick on a sheet of router matting. This router matting will prevent damage to the bowl when reversing. It's a great surface and has excellent adhesion as well as a nice soft texture

3 Once the glue has dried you can cut away any waste matting keeping any offcuts for other holding jobs

4 Now everything is ready to connect to the chuck using the faceplate ring

5 As you can see here I'm finishing a sycamore (*Acer pseudoplatanus*) platter. Bring the tailstock up, centre the bowl and add gentle pressure, turning the bowl by hand to eyeball for concentricity. Ensure all of your locking handles have been tightened up on your tailstock and start the lathe at its lowest speed then gently turning the lathe up to a workable one

6 Everything is now in position and you are ready to turn off the remaining foot to leave the design that you want. You will have to leave a small spigot where the tailstock centre lies but this can be carefully nipped off with a carving chisel. Then using a small 50mm sanding disc held in the chuck or cordless drill sand the bottom to a finish ●

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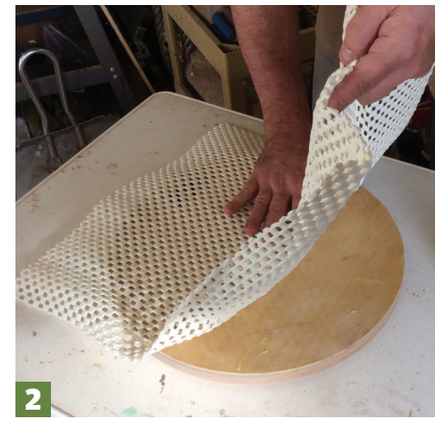
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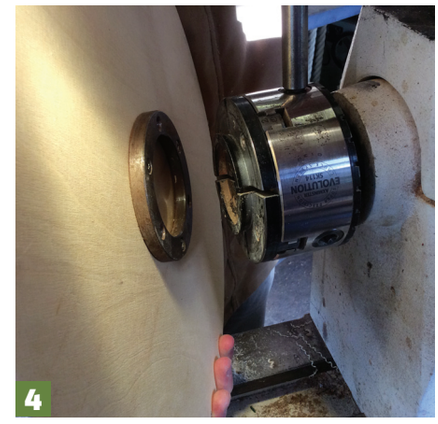
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MORE QUICK WAYS TO REVERSE AWKWARD SHAPES

In the picture above I'm reversing a poplar (*Liriodendron tulipifera*) natural edge bowl onto a wooden dome drive turned from a piece of scrap timber. Make sure you soften the surface with a piece of tissue or rag to avoid damaging the surface

Again, in this picture, I am finishing off a small walnut (*Juglans nigra*) hollow form using a similar drive leaving a small spigot where the tailstock is which can be cleaned up later with a carving knife and small sanding disc

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